

Deep Space Cargo Recovery

A 3D space simulation and precision recovery game developed with Three.js and Vite. Pilot an exploration probe, manage solar battery levels, navigate orbital hazards, and recover precious energy crystals to transport them safely back to the mothership cargo bay.

To access directly to the game, click the following link:

https://sapienzainteractivegraphicscourse.github.io/final-project-space_groovers/

Features & Highlights

Game Modes:

Testing the Probe (Standard): Free exploration with standard battery drain, unlimited time, and clear flight corridors.

Time Attack: 3-minute mission countdown with static orbital satellite debris functioning as destructive hazards.

Spinning out of Space: Hardcore mode featuring 8 actively orbiting satellites, floating micro-asteroid fields, a 90-second base timer, and a +20s Time Bonus per delivered crystal.

Controls Reference

Input Key / Action

Function

W / S

Forward / Backward Thrust

A / D

Roll Left / Right

Left / Right Arrows

Yaw Left / Right

Up / Down Arrows (or Q / E)

Pitch Up / Down

R

Extend / Retract Robotic Arm

G

Grab / Release Crystal

P

Deploy / Fold Solar Panels (Recharge)

C

Toggle Cockpit (First-Person) / Third-Person Camera

Left Click + Drag

Orbit Camera View

Mouse Wheel

Zoom In / Out

Tech Stack

Core Engine: Three.js (WebGL 3D Rendering)

Animation & Interpolation: @tweenjs/tween.js

Build Tool & Dev Server: Vite

Debug Interface: lil-gui

Getting Started

Prerequisites

Ensure you have Node.js (v16 or newer) installed on your system.

Installation & Run

If you want to enter the game through local download, follow the steps reported below:

Clone or extract the repository into your local workspace:

Bash

```
git clone https://github.com/samueleCostantinopoli/IG-Project.git
```

```
cd IG-Project
```

Install all required dependencies:

Bash

```
npm install
```

Launch the local development server:

Bash

```
npm run dev
```

Open the displayed local address (usually <http://localhost:5173>) in any WebGL-compatible modern browser.

Project Structure

Plaintext

```
├── public/
│   ├── background-music.mp3
│   └── textures/
├── src/
│   ├── components/
│   │   ├── CargoShip.js
│   │   ├── DataGalaxy.js
│   │   ├── Lights.js
│   │   └── SpaceProbe.js
│   ├── ui/
│   │   └── UserInterface.js
│   └── main.js
├── index.html
├── package.json
└── README.md
```